

Cultivating Connections: A Tangible Tabletop Game for Older Adults' Social Engagement

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Abstract—Mitigating social isolation among older adults has emerged as a critical challenge in Human-Computer Interaction. Tangible tabletop gameplay, as an accessible alternative, can promote natural social participation while reducing cognitive load. However, there remains a lack of empirical studies evaluating the effectiveness, user experience, and social impact of this technology among older adults. To provide such empirical insights, we designed and developed a co-located tangible tabletop game prototype. We evaluated with 24 older adults, employing a mixed-method approach that used traditional and digital modalities as baseline references. Video analysis and coding of 1731 social events revealed that physical affordances effectively amplify emotional resonance and trigger active self-regulation. We further proposed design considerations for tangible interactive tabletop games, such as balancing competition and cooperation, intuitiveness and operability, and multimodal interaction in top-down views. This work highlights the psychosocial potential of TUI-based multiplayer games and identifies opportunities to improve social engagement and overall well-being among older adults.

Index Terms—Tangible Tabletop Game, Tangible User Interfaces, Older Adults, Social Engagement

I. INTRODUCTION

Social engagement is an essential process for maintaining the physical and mental health of older adults, as it significantly reduces the risk of cognitive decline and enhances subjective well-being [1]. In this context, social games serve as a multifaceted shared activity that facilitates diverse social roles and interactions, leveraging a core design centered on enabling partakers to communicate, socialize, and spend time together [2]. Yet, despite this potential, the transition to digital games introduces challenges, as many older adults find conventional digital interfaces complex and unintuitive, leading to social

isolation rather than connection [3]. Furthermore, current assistive technologies inadvertently position older adults as passive recipients of care, undermining their autonomy and self-esteem [4]. This highlights a critical need for interaction paradigms that are not only accessible but also empowering and socially enriching.

Tangible User Interfaces (TUIs), which merge physical objects with digital information [5], have a unique opportunity to engage the senses and provide a more embodied experience of digital systems. For older adults, TUI technology leverages resilient sensori-motor skills and familiar physical interactions, making it more intuitive and lowering the cognitive barrier [6]. Additionally, certain features of TUI, such as equal access and embodied facilitation, can effectively be used to provide augmented social interaction experiences for seniors [7]. This potential is particularly potent when applied to tangible interactive games, which combine the playful engagement of traditional board games with the dynamic capabilities of digital systems, like IncreTable integrating real and virtual game pieces to solve in-game challenges [8]. However, existing research on TUIs for older adults predominantly focuses on individual usability and task performance, overlooking the complex group dynamics and interpersonal structures that emerge during collaborative play. Consequently, there remains a critical lack of empirically-grounded knowledge regarding how specific TUI design features influence the quality of social engagement and wellbeing in this demographic.

To provide empirical insights, we developed a co-located tangible tabletop game prototype tailored for older adults. We conducted a mixed-method evaluation with 24 participants, comparing this TUI modality against traditional card and digital multi-touch baselines. By systematically coding participants' social behaviors, we translated these findings into concrete design requirements to promote social connection

This research was supported by the National Social Science Fund of China (Grant No. 24BXW081). We also acknowledge the support of the Tsinghua University Initiative Scientific Research Program.

and mitigate interaction barriers. Our research questions are as follows:

RQ1: How can tangible tabletop games be designed to naturally foster social engagement among older adults?

RQ2: How does the tangible interaction modality influence older adults' social behaviors compared to traditional and digital baselines?

RQ3: What design considerations can be derived to guide the development of tangible tabletop multiplayer games for older adults' wellbeing?

To address these questions, we designed and evaluated a novel tangible tabletop system. Our contributions include: (1) Farmers vs. Landlord, a co-located prototype facilitating intuitive socialization and lowering cognitive barriers for older adults; (2) an empirical evaluation across different modalities demonstrating psychosocial potential of TUI in enhancing social connections; and (3) actionable design considerations for future aging-centered tangible interventions.

II. RELATED WORK

A. Games for Older Adults' Social Engagement

Traditional tabletop games have long been recognized as critical interventions for maintaining cognitive vitality and combating social isolation among older adults [2]. However, these purely analog activities lack the dynamic interactivity and real-time feedback mechanisms of modern computing. Although digital games offer such computational capabilities, they frequently introduce new barriers: complex graphical interfaces and abstract control schemes can strain older adults' working memory and attention, leading to frustration rather than enjoyment [9]. Consequently, older adults tend to prioritize games with low cognitive overhead and explicit social scaffolding [10], creating a need for a hybrid approach that balances familiarity with digital enhancement.

B. Tangible User Interfaces for Tabletop Gameplay

TUIs have emerged as a promising bridge, merging the rich social affordances of physical games with the dynamic capabilities of digital systems [11]. By leveraging older adults' sensori-motor skills and familiarity with physical manipulation, TUIs significantly reduce the cognitive load associated with digital interfaces [12]. Moreover, the shared physical space of a tangible tabletop naturally supports co-located collaboration. Features such as visibility of actions and simultaneous manipulation of artifacts provide a transparent medium for social negotiation, allowing players to focus on each other rather than the interface [7]. Despite these theoretical advantages, empirical research on how specific TUI designs influence in-depth social engagement and overall well-being remains limited. Most evaluations focus primarily on usability metrics or individual task completion, necessitating further empirical evaluation of multiplayer tangible experiences.

C. Social Interaction Patterns in Older Adults' Gaming

Unlike younger demographics, older players typically prioritize emotional support and social connection over competition in gaming contexts [13]. Within these shared tabletop

activities, specific behavioral patterns, such as role allocation, cooperation, and group decision-making, significantly shape the overall social experience. Prior empirical studies have identified several key patterns that foster positive engagement. For instance, research by Zhang indicates that flexible role assignment can greatly enhance game attractiveness and active participation among seniors [14]. Furthermore, older adults tend to derive greater enjoyment from gameplay that balances competition with explicit cooperative mechanics, as shared goals often strengthen social relationships [15]. Group cohesion is another critical pattern. Highly cohesive groups are more likely to exhibit supportive behaviors, inclusive communication, and positive social interactions during play [16]. To systematically quantify these complex interactions, we draw upon the structural lens of group dynamics [17]. This perspective allows us to decompose fluid behaviors into observable dimensions, specifically interaction patterns, role-based influences, and conflict resolution, to assess how different game modalities shape the depth and quality of older adults' social connections.

III. SYSTEM DESIGN: FARMERS VS. LANDLORD

A. Formative Insights and Design Goals

To address RQ1 and ensure our system is aligned with the specific cognitive and physical needs of older adults, we conducted a formative iterative design process involving eight seniors (4 males and 4 females, all aged over 60). Instead of proceeding immediately to digital implementation, we utilized a three-phase prototyping approach to identify interaction barriers and refine our design requirements.



Fig. 1: Static paper game prototypes. (a) Static paper prototypes. (b) Multi-touch game prototype.

1) *Iterative Prototyping Insights:* We initially constructed a paper prototype (Fig.1a) centered around a culturally familiar farming theme. While the theme resonated well and participants easily grasped the structured, turn-based mechanics, the manual tracking of game states and score calculations quickly induced cognitive overload. To automate these processes and provide dynamic visual feedback, we transitioned to a digital multi-touch prototype (Fig.1b) using an isometric camera view to prevent spatial disorientation. However, this phase revealed two critical physical barriers: (1) tactile ambiguity, as participants struggled with the lack of haptic confirmation on the smooth screen, and (2) fine motor fatigue, as continuous drag-and-drop gestures proved unintuitive and tiring. These insights confirmed that neither paper prototype nor digital

interfaces were sufficient, highlighting the necessity of the TUI that bridges physical intuition with digital scaffolding.

2) *Design Goals*: Based on these formative insights, we outline four actionable design goals (DGs) to answer RQ1 and guide the development of our TUI game:

a) *DG1: Enhancing Playfulness and Familiarity of Tangible-Digital Representation*: To minimize learning costs and enhance game acceptability [12], [18], the system should employ physical tokens and familiar thematic narratives (e.g., farming) that leverage natural, embodied interactions rather than abstract digital metaphors.

b) *DG2: Achieving Effectiveness and Balance of Cooperative and Competitive Dynamics*: To promote social interaction and rapid trust-building among unfamiliar players, the game must balance mild competition with shared objectives. By employing “win together or fail collectively” mechanics [19], the system effectively prevents social exclusion and naturally encourages verbal negotiation.

c) *DG3: Provide Dynamic Digital Feedback to Sustain Flow*: To sustain immersion and a flow state for natural social interaction, the system must automate rule enforcement and game state tracking. By shifting complex memory and calculation tasks to the background, the system relieves players from administrative burdens, thereby preventing cognitive fatigue. Concurrently, as the tracking system detects valid physical manipulations, it instantaneously triggers real-time positive audio-visual feedback [20], offering a continuous sense of achievement to motivate active participation.

d) *DG4: Maintaining the safety of social interactions within controlled boundaries*: To accommodate physical limitations and prevent fatigue, interaction gestures must avoid repetitive strain within carefully paced sessions. Furthermore, to mitigate potential multiplayer conflicts [21] and maintain safe social boundaries, the system must enforce transparent and fair rules that naturally prevent misunderstandings.

B. Game Design: Farmers vs. Landlord

Guided by our established design goals, we developed Farmers vs. Landlord, a turn-based tangible tabletop game designed for four players. The system utilizes a mapping mechanism that links physical building blocks on a capacitive touchscreen to a digital virtual environment. Drawing inspiration from popular Chinese tabletop games, we incorporated the 3-vs-1 asymmetric role dynamics of “Dou Di Zhu” (Fight the Landlord) and the distinct private or public zone spatial arrangements found in Mahjong. The game aims to balance cooperation and competition while exercising players’ strategic thinking and fine motor skills in a familiar, engaging setting.

1) *Core Mechanics and Tangible Interaction (Addressing DG1)*: Farmers vs. Landlord (Fig.2) involves three “farmers” and one “landlord” developing their respective homesteads. After a simple rock-paper-scissors to determine roles, players use physical house-shaped building blocks to construct their homes on the tabletop, which are instantly mapped into the virtual space.

Game Rules — Farmer

Task1: Harvest 10 crop blocks and successfully defeat the landlord.



①Crop Block Rotation
Full rotation for crops.

②Click on the crops to
harvest and redeem crop

Game Rules — Landlord

Task1: Build roads and drive to collect taxes from the farmers.



①Double-click to lay a road
Proceed to the farmer's home.

②Place a vehicle until
reaching the house.

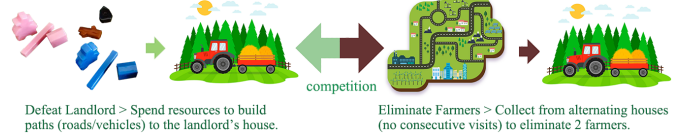


Fig. 2: Farmer and landlord rules in the game

a) *Farmers’ Rules*: Farmers are equipped with custom tangible crop blocks. To harvest, players must rotate a tangible token to a specific angle when the crop appears on-screen, and then drag it to their virtual homestead (Fig.3). Harvested crops yield physical resource blocks. This rotation and dragging naturally exercises hand-eye coordination. Once farmers accumulate sufficient assets, the system recommends strategic upgrades, such as purchasing vehicles or constructing roads.

b) *Landlord’s Rules*: The landlord faces a more strategically demanding task that requires disrupting farmers’ development while executing road construction plans. To execute this, the landlord utilizes an asymmetrical bimanual control interface, where the right hand manages forward propulsion, and the left hand dictates steering, thereby demanding complex bilateral motor coordination.

c) *Modular Defense and Disasters*: To introduce dynamic challenges, natural disasters occur randomly. Players must use modular building blocks to construct physical barriers against these digital threats. To achieve this, the system employs an overhead depth sensor to continuously monitor the tabletop’s topological data. The physical width and height of the stacked blocks are calculated as real-time volumetric input and mapped into the 3D visual space [22]. Once the detected physical volume surpasses a pre-defined digital threshold, the system automatically registers the disaster as mitigated. Faster players can assist slower ones, providing an opportunity for cross-role cooperation.

2) *Gameplay Rhythm and Social Dynamics (Addressing DG2)*: To scaffold balanced social dynamics, the game employs a strict turn-based system. Each player is allocated 45 seconds per turn, providing ample time for observation and strategic planning without inducing time-pressure anxiety [23]. Additional pauses are integrated after each action for resource exchange, introducing a clear concept of “initiative” [19]. This explicitly structured agency signals when players should negotiate, preventing chaotic overlapping communication and ensuring every participant has a guaranteed window to voice their strategies. The 3-vs-1 asymmetric design forces the three farmers into a tightly aligned cooperative task. Conversely, the landlord faces a challenging competitive environment, enhancing the overall excitement.

3) *Multisensory Feedback and Cognitive Offloading (Addressing DG3)*: To sustain player immersion and flow, the design caters to the specific sensory needs of older adults [24]. We addressed the visual obscurity of traditional top-down screens by converting 2D digital elements into touchable 3D forms. Information clarity is enhanced through dual encoding and high-contrast, enlarged UI elements. Furthermore, the system leverages background computation to handle all score-tracking. Real-time, positive audio-visual feedback is triggered upon successful actions, offering a continuous sense of achievement that motivates repeated cooperation.

4) *Conflict Mitigation and Game Pacing (Addressing DG4)*: To maintain social interactions within safe boundaries, we implemented specific control measures. First, the mechanics are explicitly rigorous and fair. Players can only harvest assets when they simultaneously meet the precise criteria for rotation angle and drag location, preventing disputes over scoring. Second, the turn-based waiting periods naturally encourage off-turn players to supervise and assist the active player, maintaining a collaborative atmosphere. Finally, to prevent physical fatigue, the overall game duration is strictly capped at 15 minutes. If no winner emerges, the system automatically applies a secondary judgment rule based on accumulated resources to declare a victor.

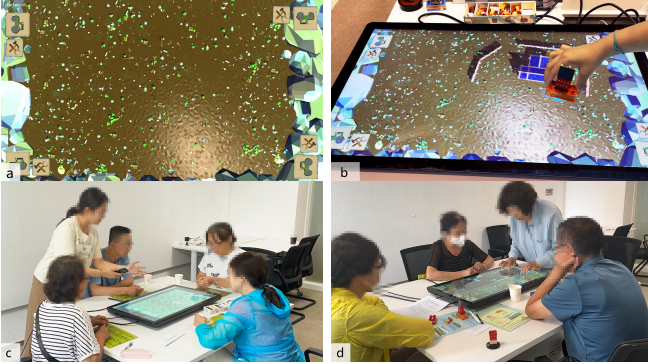


Fig. 3: Interaction process. (a) Initial scene. (b) Dragging the house block.

5) System Implementation:

a) *Hardware*: We use a method for generating and optimizing triangular patterns for tangible surface tokens on capacitive touchscreen [25]. The method uses spatial marker data to form patterns and employs a template-matching algorithm for identification. Poisson disk sampling is applied in the feature vector space with constraints to ensure patterns meet size, shape, and distance requirements. The method improves tangible objects recognition by addressing issues with capacitive token designs and offers more unique patterns. To support the volumetric tracking of stacked blocks, an overhead Azure Kinect depth camera is securely mounted above the interactive surface. Additionally, we create a series of physical tokens in various shapes and sizes, tailored to accommodate older users' operating habits and different types of digital content.

b) *Software*: We used the Unity2021.3.17f1 Engine to develop the game Farmers vs. Landlord, which was created on URP and focused on Tangible and Multi-touch screen platforms. The game included a menu for planting crops, a road construction system, growing and harvesting, as well as a camera control system. Objects are painted with vertex color and UVs for different model parts.

IV. METHODOLOGY

To answer RQ2 and mitigate older adults' technology anxiety, we employed a scaffolded design to ecologically assess the tangible tabletop's unique social impact on older adults, establishing behavioral baselines via traditional card and digital multi-touch modalities.

A. Participants

Following the United Nations standards and the standards established by the HCI community for older adults [26], we recruited a total of 24 older adults between the ages of 60 and 75 (mean age 67.87 ± 2.86 years, 10 males and 14 females) from local community centers and senior activity groups for our study. A wide range of educational backgrounds was represented, and more than half of the participants had a professional degree. Inclusion criteria required participants to have normal or corrected-to-normal vision and hearing, and no severe clinical cognitive or motor impairments that would preclude gameplay. We provided the screened, eligible, and interested older adults with an informed-consent document and a demographic questionnaire with the experimental purpose and task. Then, we randomly assigned participants to six groups of four. After the experiment, participants received a monetary compensation of 70 CNY for their contribution. The entire study was conducted in our laboratory and received approval from our university's ethics and morals committee.

B. Procedure

When participants arrived for their experiment session, researchers first explained the study's purposes and provided standardized instructions to familiarize them with the operation of both the physical props and the digital screen.

In accordance with our progressive baseline design, the experimental phases transitioned from the most familiar modality to the most novel. This intentional sequence functioned as an essential warm-up, allowing participants to establish their natural social dynamics in a low-stress environment before encountering the novel TUI. Consequently, they could focus entirely on the interaction modality rather than struggling with basic rules or social apprehension. To mitigate potential order effects, we incorporated mandatory 15-minute washout periods between sessions to minimize fatigue and emotional carryover. The progressive phases were structured as follows:

1) *Phase 1: Traditional Card Baseline*: The participants played a poker-based card game (Doudizhu) without digital intervention, in which three players bid competitively to determine a landlord, who then played against the other three farmers. Players took turns playing valid card combinations with the goal of emptying their hands using a standard deck.

2) *Phase 2: Digital Multi-touch Baseline:* Participants played a multi-touch screen game that replicated the original rules but transitioned all interactions to a digital interface. This phase served to explore the impact of virtual screen interactions on their social dynamics.

3) *Phase 3: Tangible Tabletop Intervention:* Participants engaged with our novel tangible tabletop game, which adapted the core cooperative and competitive elements into a format combining multi-touch and physical object interactions. This condition was designed to investigate the unique supportive role of tangible interaction in fostering social connections.

Following each session, participants completed a modality-specific questionnaire during their washout break. After all three phases, they filled out a comparative survey and engaged in a 25-minute semi-structured interview.

To minimize observer bias, two researchers monitored the sessions from a diagonal distance of approximately two meters outside the participants' visual field, providing only necessary technical and safety support. Furthermore, synchronized dual-camera arrays were positioned diagonally to continuously capture both the participants' social behaviors and the dynamic tabletop game state for post-hoc blind video coding.

C. Data Collection and Measures

TABLE I: Video Behavioral Coding Framework

Modules	Contents	Details
Interaction patterns	Verbal and non-verbal communications (Multiplayers)	Verbal: unidirectional and multi-directional Non-verbal: body language, expression, eye contacts
Role influence & status change	Roles 1: Landlord Roles 2: Farmers	The specific behaviors exhibited by participants when assigned or changing roles
Conflict and resolution	Task-related or relationship-based conflicts	Conflicts: game missions or relationships Methods: Compromise / Confrontation
Individual behaviors	Personal behaviors (One player)	Personal facial expressions, physical actions, emotions, and verbal speaking
Others	Other unclassified behaviors	

Post-game experiences were assessed using the Game Experience Questionnaire (GEQ) [27], a 5-point Likert Social Connectedness Scale (SCS) [28], and adapted items (6-10) from the Collective Efficacy Scale (CSC) [29]. Furthermore, we captured 782 minutes of dual-camera gameplay video. Audio was transcribed via AI within MAXQDA and manually verified. We then applied a deductive thematic analysis [30] grounded in group dynamics [17] to code and interpret observable social behaviors across modalities.

Researchers established a unified video coding framework, classifying potential interactive behaviors into three primary modules (Table.I): interaction patterns, role influence with status changes, and conflict generation and resolution, along with a separate module for individual behaviors. To minimize individual bias and ensure alignment, four researchers, all

of whom are co-authors, initially cross-checked a randomly selected 20% subset of the audio-video recordings and transcripts, resolving discrepancies through intensive discussion until full consensus was reached. The remaining videos were then independently coded according to this finalized framework. In total, 1731 social events were identified, comprising 378 behaviors from the card game, 576 from the touchscreen game, and 777 from the tangible game.

V. RESULTS

This section presents empirical findings on how the system impacts older adults' social interactions and evaluates behavioral differences across modalities. These results provide the foundational evidence to address RQ3 regarding the design considerations detailed in the Discussion.

A. Descriptive Statistics

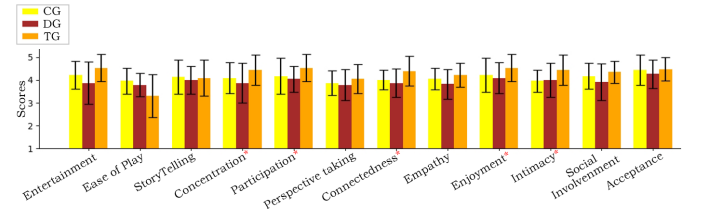


Fig. 4: The scores of items in the GEQ, SCS, and CSC scale for three different game conditions: CG, DG, and TG.

All scale data passed normality tests. We conducted a one-way repeated measures Analysis of Variance (ANOVA) to evaluate differences across the three progressive modalities: traditional card game (CG), digital multi-touch game (DG), and tangible tabletop game (TG) (Fig.4). For all significant main effects, Bonferroni-corrected post-hoc pairwise comparisons were performed to determine specific group differences.

1) *Game Experience and Engagement:* TG ($M = 4.52, SD = 0.60$) yielded higher enjoyment than CG and DG ($F(2, 46) = 4.85, p = .012$). Similarly, concentration was highest during TG ($M = 4.41, SD = 0.68; F(2, 46) = 3.51, p = .039$). Frustration and social presence showed no significant differences across conditions.

2) *Social Connectedness and Intimacy:* TG elicited the highest user participation ($F(2, 46) = 3.42, p = .043$). Crucially, TG fostered stronger social connectedness ($M = 4.39, SD = 0.66; F(2, 46) = 6.82, p < .005$) and intimacy ($M = 4.43, SD = 0.66; F(2, 46) = 6.55, p < .005$), even among unacquainted participants. No significant differences emerged for perspective-taking, social satisfaction, or empathy (all $p > .05$).

3) *Acceptance and Methodological Validation:* TG showed a marginal advantage in encouraging future social participation ($F(2, 46) = 3.07, p = .054$), while general game acceptance did not differ. Notably, TG scored lowest in "ease of play" ($M = 3.22, SD = 0.90$) versus CG ($M = 3.91, SD = 0.60$) and DG ($M = 3.74, SD = 0.91$). This empirically validates our scaffolded procedural design: introducing TG modality last

prevented initial technology anxiety while maximizing final social engagement.

B. Video Behavioral Analysis

1) *Communication and Conflict Resolution*: To systematically interpret the 1,731 coded events, we conducted a thematic analysis across modalities. Verbal communication varied significantly (Fig.5). While all conditions facilitated analogical learning (e.g., P3: *Is it similar to 'Zhuang Shang You'?*), TG uniquely sustained engagement across all phases. Its physical execution created natural pacing intervals, promoting casual conversations over passive waiting. Furthermore, modalities transformed conflict resolution. DG's simultaneous touching caused disruptive task-induced conflicts, whereas TG conflicts were primarily spatial and strategic (e.g., P10: *We're supposed to cooperate, but stay out of my space.*). Moreover, minor TG physical mishaps (e.g., misplacing a token) triggered collective laughter, naturally diffusing tension and fostering cohesion.

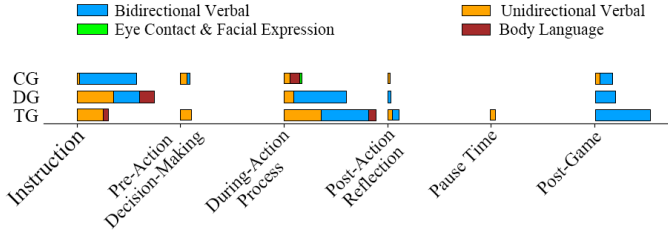


Fig. 5: Distribution of social interactions across progressive game phases. Bar widths represent the relative frequency of observations within each categorical phase.

2) *Role Dynamics and Strategy Emergence*: While game roles shaped initial alliances, interaction modalities dictated their evolution. Farmers predictably formed survival alliances, such as P10 urging a partner: *“You need to keep me safe from him.”* Conversely, Landlords exerted competitive pressure, with P4 warning: *“If you earn too much, I’m coming to raid your home!”* However, TG uniquely fostered implicit competition within alliances and unexpected support across adversaries. Its visible physical assets prompted discreet comparisons, recognizing a deficit, P8 privately practiced single-handed rotations to accelerate harvesting. Unlike the strict oppositional behaviors in baselines, TG facilitated cross-role prosociality, demonstrated when Farmer P19 gesturally guided Landlord P18’s driving. This highlights how shared tangible spaces act as social catalysts, softening competitive roles into fluid, supportive interactions.

3) *Emotional Engagement and Self-Regulation*: The behavioral distribution (Fig.6) revealed that tangible interaction significantly expanded participants’ emotional bandwidth. While the CG and DG baselines elicited calm or relaxed states, TG evoked a highly dynamic emotional spectrum. Although strategic gameplay inherently introduced pressure, TG’s physical affordances amplified these expressions, leading to frequent surprise, urgency, laughter, and frustration. For example, a Landlord (P4) grinned after being strategically

blocked by a Farmer, indicating an immersive enjoyment of tactical pressure. Crucially, TG’s physical and temporal constraints frequently triggered audible *self-talk*, a self-regulatory mechanism rarely observed in the baselines that helps older adults alleviate cognitive tension [31]. For instance, facing a tangible time limit, one participant muttered: *“I’ve only managed one... Oh, time’s up.”* This highlights how TG engages older adults in active cognitive pacing, transforming task-induced stress into a manageable and engaging experience.



Fig. 6: Distribution of individual behaviors across progressive game phases. Bar widths represent the relative frequency of observations within each categorical phase.

C. Qualitative Insights from Post-Game Interviews

1) *Perceptions of Tangible Interaction*: First-time TUI users found the modality highly engaging. As P2 noted, *“It makes you want to keep interacting... even if you don’t always succeed.”* Participants found physical manipulation more immersive and cognitively stimulating than touch inputs, praising the required hand-eye coordination. Furthermore, physical resource accumulation enhanced psychological ownership, with one user remarking, *“The more I collect, the happier I feel.”* However, novel physical affordances presented an initial learning curve. Consequently, participants suggested scaffolded difficulties (e.g., P21: *“levels one, two, three”*) to optimize older adults’ learning experiences.

2) *Subjective Experiences of Social Dynamics*: Compared to conventional activities, games as a social medium provided stronger interactivity and engagement. Participants highly valued how the natural pacing of the tangible game provided comfortable spaces for broader social connection; the shared physical environment allowed in-game interactions to seamlessly transition into daily conversations. The interplay of competition and cooperation was particularly well-received. Participants actively relished the *“tension”* and *“excitement”* of competition, with P15 admitting a thrilling *“fear of losing”* which P2 described as distinctly *“different from other social activities.”* While cooperative mechanics successfully elicited mutual assistance, the engaging competition was consistently highlighted as a unique driver of deep social immersion.

VI. DISCUSSION

Our empirical findings reveal how specific tangible mechanics shape the social dynamics of older adults. To address RQ2 and RQ3, we synthesized these insights into actionable design implications for aging-centered tabletop games, followed by

a reflection on how TUIs reframe the measurement of social engagement.

A. Design Implications for Aging-Centered TUIs

a) *D11: Leveraging Embodied Affordances for Non-verbal Scaffolding:* Video analysis revealed that participants (e.g., P5, P8) frequently used game pieces not just for task completion, but as non-verbal cues, such as pointing with a block to initiate a suggestion or sliding a shared component to signal a collaborative intent. This suggests that TUIs should be designed to support observable physical actions, which reduce the cognitive effort required for verbal negotiation and reinforce group cohesion through shared spatial awareness.

b) *D12: Facilitating Sensory Accessibility through Multimodal Feedback:* Our results indicate that the "tangible-digital" mapping triggers reminiscence and personal narratives. For example, a participant (P3) noted positive reminiscence: "Harvesting crops reminds me of picking fruits in the orchard". Combined with our questionnaire data, we found a strong correlation between perceived tactile realism and social connection. Thus, designers should select culturally familiar physical metaphors that act as emotional catalysts, transforming a functional game task into a meaningful shared life experience.

c) *D13: Balance in Hybrid Asset Accumulation:* A critical insight emerged regarding the sense of achievement. While participants appreciated the fairness and precision of digital counting, they expressed a higher sense of acquisition when counting physical building blocks. This behavioral pattern suggests a "Hybrid Asset Model": utilizing physical accumulation to provide immediate, tactile satisfaction of progress, while employing digital overlays to ensure transparency and prevent interpersonal conflicts arising from manual tallying errors.

d) *D14: Supporting Interaction Flexibility and Spatial Accommodation.:* Driven by varying levels of physical dexterity and comfort, older adults exhibit highly diverse interaction styles, transitioning between 2D touch and 3D physical manipulation. Furthermore, tangible tabletops inherently create a shared spatial context that allows for both public actions (e.g., reaching across the table) and private arrangements (e.g., organizing personal tokens). Therefore, interfaces must offer interaction flexibility, enabling older players to manipulate objects using whatever idiosyncratic gestures or personalized strategies feel most natural to them, much like the freedom of holding and sorting traditional playing cards.

e) *D15: Injecting Desirable Difficulties" to Stimulate Social Scaffolding:* While TUI design traditionally emphasizes intuitive affordances [32], our findings suggest a delicate balance is needed. Overly simplistic operations fail to sustain engagement. Conversely, introducing slight operational complexities (e.g., navigating the physical car token) acted as a desirable difficulty." This mechanical friction functioned as a social scaffold, actively stimulating peer-teaching, mutual assistance, and shared laughter. Designing these mild, collaborative challenges can transform operational hurdles into rich opportunities for social interaction and cognitive exercise.

B. Reframing Social Engagement in Tangible Gameplay

While existing literature has extensively conceptualized team processes through the lens of continuous, non-monolithic dynamics [33], [34], our study offers a novel methodological operationalization of these theories for old collaborative settings. Unlike screen-focused systems, TUIs leverage embodied cognition to capture nuanced social behaviors, including object-mediated communication, physical negotiation, and collective space management, which are typically omitted in digital logs. Addressing the potential risk of social isolation for the Landlord in the 1-vs.-3 asymmetric design, our qualitative data revealed a dynamic balance rather than exclusion. Specifically, 19 responses highlighted the positive tension of competition, with P15 noting, "There's a sense of tension in the competition process, afraid of losing," while 16 emphasized cooperation. This mechanical competition was profoundly softened by the physical interaction and hand-eye coordination, which participants found uniquely enriching, as P2 stated, "The feeling of competition is different from other social activities." Ultimately, the physical tokens transformed competitive pressure into a safe, shared social catalyst, fostering spontaneous cross-faction mutual assistance and peer-teaching that effectively reframed the asymmetric conflict into a collaborative experience, as P14 summarized: "I find the real value in helping and learning; winning or losing does not matter."

Crucially, while our progressive baseline design functioned as a scaffolded training process that naturally lowered social apprehension over time, the distinct form of these behaviors was strictly tied to the tangible modality. Even as participants grew more familiar with one another, pro-social behaviors, such as physically sharing assets, were structurally precluded in the digital or analog baselines. Furthermore, the ability of tangible interactions to trigger nostalgic reminiscence (e.g., linking physical harvesting to life stories) lowered entry barriers and fostered emotional resonance. Ultimately, this multimodal engagement of touch, sight, and sound transcended traditional gameplay to create an inclusive and emotionally supportive social environment. These spatially embedded metrics suggest that future evaluations of aging-centered designs must look beyond digital performance scores and quantify the physical, object-mediated social interactions.

C. Limitations and Future Work

Our study presents several limitations. First, to prioritize psychological comfort and ecological validity, we employed a progressive baseline rather than a strictly counterbalanced design. While this successfully mitigated initial technology anxiety, it limits our ability to completely rule out order and learning effects. Second, the segmentation of game phases constrained data collection during inter-round intervals, potentially truncating richer social interactions. Finally, although our video-based analysis utilized cross-checked coding, manual qualitative approaches remain susceptible to coder bias.

Future research should integrate computational methods to objectively quantify physical social engagement metrics. Addi-

tionally, while this study establishes a baseline for older adults aged 60-75, subsequent work must investigate older cohorts (75+) facing distinct cognitive constraints. Exploring how group size, gender dynamics, and prolonged session lengths affect psychosocial outcomes will further illuminate the long-term cognitive and social benefits of tangible gameplay.

VII. CONCLUSION

This paper presented a mixed-methods study investigating how tangible tabletop gameplay fosters social engagement among older adults. By designing and evaluating the tangible game, *Farmers vs. Landlord*, our group-dynamics analysis empirically demonstrates how physical affordances uniquely amplify emotional resonance and trigger active self-regulation. Translating these insights, we proposed actionable design implications, including the hybrid assets and the strategic use of desirable difficulties to optimize co-located play. Ultimately, this work highlights the transformative potential of tangible interactions in fostering relational play within aging-centered design, providing a robust empirical and practical foundation for mitigating social isolation through inclusive, multisensory technologies.

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